

# Ali Shirzad

Software Engineer | Python | LLM integration | Docker | Volvo Group, Chalmers

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## SUMMARY

Software Engineer with a Master's in Complex Adaptive Systems and 1+ years of automotive software development experience at Volvo Group Trucks Technology, where I develop automation tooling and machine learning solutions for vehicle dynamics. A Python tool I created saves the team about **400 engineering hours a year** and is actively used in production; my thesis model forecasts truck vibration in **seconds** versus conventional simulations that take hours to days. On my own time I build projects end to end, from a full-stack analytics web app to machine learning on vehicle data.

## WORK EXPERIENCE

**Software Engineer Intern** | Volvo Group Trucks Technology (GTT) 06/2026 – 09/2026

*Tools: Python, XML, Excel VBA, automated testing*

- Built a Python tool that generates pinout (wiring) documentation engineers wrote by hand
- Extended an internal Python configuration tool to a second family of connectors
- Delivered it alone: Python script, drawing template, XML files, validation, tests, logging

**Master's Thesis Student** | Volvo Group Trucks Technology (GTT) 01/2026 – 06/2026

*Machine Learning-Based Model Order Reduction for Full Vehicle Finite Element Models*

*Tools: Python, PyTorch, MATLAB, finite element (FE), time-series data*

- Built and trained neural network models that predict truck vibration in seconds, not hours to days
- Made the models small enough to run later on a data logger beside live sensors
- Led the work from start to finish and evaluated models based on physical fidelity, not just training error

**Software Engineer Intern** | Volvo Group Trucks Technology (GTT) 06/2025 – 08/2025

*Tools: Python, automated testing*

- Automated a manual task, saving the team **400 hours/year**; the tool is now used in production
- Designed the diagram template and rebuilt the Python script that generates the documents
- Automated output validation by writing unit tests that verify generated documents against the specification

## TECHNICAL SKILLS

**Languages:** Python, C, C++, SQL (PostgreSQL), JavaScript, MATLAB

**Frameworks and Libraries:** PyTorch, TensorFlow, NumPy, Pandas, FastAPI, React, Node.js

**Developer Tools:** Git, Docker, Linux, CUDA, CI/CD, HPC and multi-GPU, XML, Excel VBA, automated testing

**Machine Learning:** neural networks, time-series modelling, surrogate and reduced-order models, LLM integration, MCP, agent workflows

**AI-Assisted Development:** Claude Code, Codex, GitHub Copilot

**Spoken:** Swedish (fluent), English (fluent), Persian/Dari (native)

## EDUCATION

**Chalmers University of Technology** | Civilingenjörsexamen (M.Sc. in Engineering) 2021 – 2026

*Master of Science (M.Sc.), Complex Adaptive Systems*

08/2024 – 06/2026

*Bachelor of Science (B.Sc.), Computer Science and Engineering*

08/2021 – 06/2024

## PROJECTS

**Local AI Coding Agent** | Python, Docker, local LLMs Personal project

- Python agent that turns a plain-English task into working code and tests it itself
- Runs each generated program in a Docker sandbox with automatic security checks

**SmartBI Insights** | React, FastAPI, PostgreSQL, REST APIs, Docker Personal project

- Full-stack analytics web app for retail suppliers, deployed with a single Docker command
- Every figure is checked against the database, so the AI cannot invent a number
- Each customer's data is kept separate, enforced inside the database itself

**GPU-Accelerated CFD Solver** | Python, CUDA, MPI, NVIDIA AMGX, A100 and H100 Chalmers

- CUDA fluid-flow simulator running across many GPUs on an HPC cluster

## REFERENCES & MORE

References available on request | More projects at [github](#)